



DELIVERY APP UPDATE · MARCH 28

# Feature 4: Live Tracking

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## What We Built Today

Three files were created to handle the tracking (status changes) for deliveries.

### File 1: DeliveryRequestRepository.java

#### What it does

- This is the app's remote control for the delivery\_requests table in MySQL.
- It lets you find, save, and update orders without writing any SQL code.
- Spring reads the method names and generates SQL automatically.

#### Simple Example

When you write `findByStatus(PLACED)`, Spring translates it to:

```
SELECT * FROM delivery_requests WHERE status = 'PLACED'
```

You never touch SQL yourself.

### File 2: TrackingEventRepository.java

#### What it does

- Every time a delivery's status changes, one row gets saved in the tracking\_events table.
- This repository lets you fetch all status changes for one order, sorted oldest to newest.
- This is the raw data that builds the tracking timeline customers see.

#### Real Example

When a customer opens the tracking page, this repository gives you:

```
10:00 – Order placed  
10:15 – Assigned to agent  
11:30 – Out for delivery  
12:05 – Delivered
```

Each line above is one row returned by this repository.

### File 3: TrackingService.java

This is the brain of Feature 4. It has three jobs:

#### Job 1 — State Machine (The Rulebook)

Decides if a status change is allowed or blocked.

- ✓ ASSIGNED → OUT\_FOR\_DELIVERY (allowed)
- ✗ PLACED → DELIVERED (blocked — skipped steps)

## Job 2 — Actually Update the Status

When an agent clicks "Mark as Out for Delivery", this runs:

1. Checks the rulebook — is this move allowed?
2. If yes → saves a TrackingEvent record + updates the order status.
3. If no → throws an error and nothing changes.

## Job 3 — Tell the Page Which Buttons to Show

Returns a list of statuses this order is allowed to move to. The page uses this to decide which buttons to show the agent. Invalid options simply don't appear.

## pom.xml Fix

Fixed three problems in the project configuration file:

1. Spring Boot version was 4.0.5 (doesn't exist) → changed to 3.3.5 (real, stable version).
2. Dependency was spring-boot-starter-webmvc (wrong name) → changed to spring-boot-starter-web (correct name).
3. Three fake test dependencies removed, replaced with one real one: spring-boot-starter-test.

These fixes let IntelliJ download the correct libraries and clear all red error lines.

## What's Next

| Date     | Task                                          |
|----------|-----------------------------------------------|
| March 29 | Build TrackingController (the web endpoints)  |
| March 30 | Build the tracking page UI (Thymeleaf HTML)   |
| March 31 | Test everything, fix issues, commit Feature 4 |